Zahra Fallah

MSc Researcher in Cognitive Science & Human Computer Interaction
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RESEARCH PROFILE

Cognitive Science MSc researcher working on VR-based behavioral experiments, embodied interaction, and perceptual decision-making. Experienced in Unity-based experimental design, computational behavioral analysis, eye-tracking, and immersive HCI research. Interested in collaborative learning, multimodal interaction, and cognitive technologies.

RESEARCH INTERESTS

- Human-Computer Interaction
- Virtual Reality
- Perceptual Decision-Making
- Embodied Cognition
- Educational Technology
- Computational Cognitive Modeling

EDUCATION

2022 – 2025 **MSc in Cognitive Science** — **University of Tehran**
Thesis: Effects of Altered Sensorimotor Feedback on Perceptual Decision-Making, Confidence, Cognitive Load, Agency, and Embodiment in Virtual Reality. **GPA: 4.0/4.0**

2014 – 2019 **BSc in Mechanical Engineering** — **University of Tehran**
Senior Project: Design and Analysis of Optimal Motor Strategies Using EMG Signal Processing to Counteract Muscular Fatigue in Archery. **GPA: 3.36/4.0**

RESEARCH EXPERIENCE

2023 – 2025 **MSc Thesis Research: Perceptual Decision-Making in Virtual Reality**

- Designed Unity-based VR experiments investigating visuomotor feedback, embodiment, confidence, and perceptual decision-making.
- Implemented hand gesture interaction and iterative sensory feedback paradigms to study adaptive behavior and cognitive workload.
- Applied Drift Diffusion Modeling and behavioral analysis in Python.

2024 **Cognitive Pathology VR Project**

- Developed VR navigation tasks embedding addiction-related cues to study cognitive and physiological responses.
- Measured spatial memory, path integration, and HRV-based cognitive load.

2023 **Agency and Automation Interaction Project**

- Designed continuous HCI tasks to study perceived agency under different automation levels.
- Collected eye-tracking and pupillometry data to examine attention and adaptive behavior.
- Analyzed behavioral and physiological data with MATLAB.

PUBLICATIONS & PRESENTATION

2025 **Effects of Altered Visuomotor Feedback on Perceptual Decision-Making, Confidence, and Embodiment in VR** — **Manuscript in Preparation**
Finalizing results for submission to top-tier HCI venues. Features a novel integration of Drift Diffusion Modeling (DDM) to quantify the cognitive costs of asynchronous visuomotor feedback.

2025 **Embodied decision-making in VR: A Computational and Metacognitive Analysis** — **Poster Award**
2nd Place Award, Basic and Clinical Neuroscience Congress (BCN 2025). Presented empirical findings on the "performance-metacognition paradox" in immersive interaction.

ACADEMIC EXPERIENCE

Present **Project Mentor, Psychology of Learning** — **University of Tehran**

- Mentoring student projects in experimental psychology and applied learning.
- Providing guidance on behavioral data analysis and experimental design.

Present **Project Mentor, IT in Psychology** — **University of Science and Culture**

- Assisting students with game-based and AI-driven experimental projects.
- Supporting study design, implementation, and analysis.

2025 **Teaching Assistant, Computational Neuroscience** — **Neuromatch Academy**

- Assisted in international computational neuroscience and modeling instruction using Python.

PROFESSIONAL & APPLIED EXPERIENCE

2025 **LLM Agents Hackathon** — **Sharif University**

- Developed LLM-based educational assistant to investigate adaptive reasoning and personalized learning.
- Designed interdisciplinary algorithmic challenges for educational contexts.

2023 – Present **Founder, Iran Minecraft Academy** — **Tehran, Iran**

- Designed game-based learning environments for programming and computational thinking.

2019 – Present **Co-founder, Crange-Ghaf Game Studio** — **Yazd, Iran**

- Created interactive board and PC games to study adaptive decision-making and collaboration.

Apr 2023 **Cognitive Specialist, Cultural Documentation Project** — **Khuzestan, Iran**

- Conducted naturalistic behavioral observations in children, contributing to a collaborative publication.

Summer 2021 **UX Research Intern, Web Rokh** — **Tehran, Iran**

- Evaluated user behavior and decision-making on e-commerce platforms using cognitive principles.

TECHNICAL SKILLS

Programming:
R, Python (NumPy, Pandas), MATLAB, C#

Methods:
VR experiments, Game-based experiments, Serious Game, cognitive modeling, DDM, Bayesian inference, behavioral analysis, UX Research

Bio-Signal Processing:
EMG (Motor Control Analysis), Eye-tracking & Pupillometry, HRV

VR & Research Software:
Unity, Psychtoolbox, Tobii Pro Lab, EEGLAB, Git

Languages:
English (TOEFL iBT: 93), Persian (Native)

HONOURS & AWARDS

2024 **2nd in cumulative GPA, MSc Cognitive Science, University of Tehran**

2022 **2nd among 1,000+ in Iranian National MSc Entrance Exam (Cognitive Science)**

2014 **756th among 2,000,000 in Iranian National BSc Entrance Exam (Math/Physics)**

2012 **3rd Place in 14th Khwarazmi International Award in Nanotechnology and National Patent Inventory for Nanotech project on oil-seawater separation**